FIUS-MoveSense: Ultrasonic Movement Detection Using Smart Ultrasonic Sensors (FIUS) Machine Learning

Conclusion Summary
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Machine Learning

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Project Overview

The FIUS-MoveSense project demonstrates an ultrasonic sensing system capable of detecting human movement using signal processing and machine learning. The system combines a Fully Integrated Ultrasonic Sensor (FIUS) with a Red Pitaya STEMlab board to acquire ultrasonic echo signals. Each scan contains approximately 50,000 samples representing the reflected ultrasonic waveform. These signals are processed and converted into meaningful numerical features that enable machine learning algorithms to classify movement patterns.

System Architecture

The developed system follows an end-to-end processing pipeline. First, ultrasonic pulses are emitted by the FIUS sensor, and the reflected echoes are captured by the Red Pitaya ADC platform. The raw signals are filtered and processed using a Bessel low-pass filter, Hilbert envelope detection, and peak detection to identify object reflections. After preprocessing, 38 features are extracted from each scan including time-domain, zone-based, frequency-domain and temporal features. These features form the input for machine learning classifiers.

Key Project Metrics

Metric	Result
Dataset Size	25,490 ultrasonic scans
Samples per Scan	~50,000 signal samples
Extracted Features	38 engineered signal features
Best Model	Random Forest Classifier
Model Accuracy	≈ 90.5 %
Real-World Test Success	9 / 10 scenarios correctly detected

Functional Diagram:

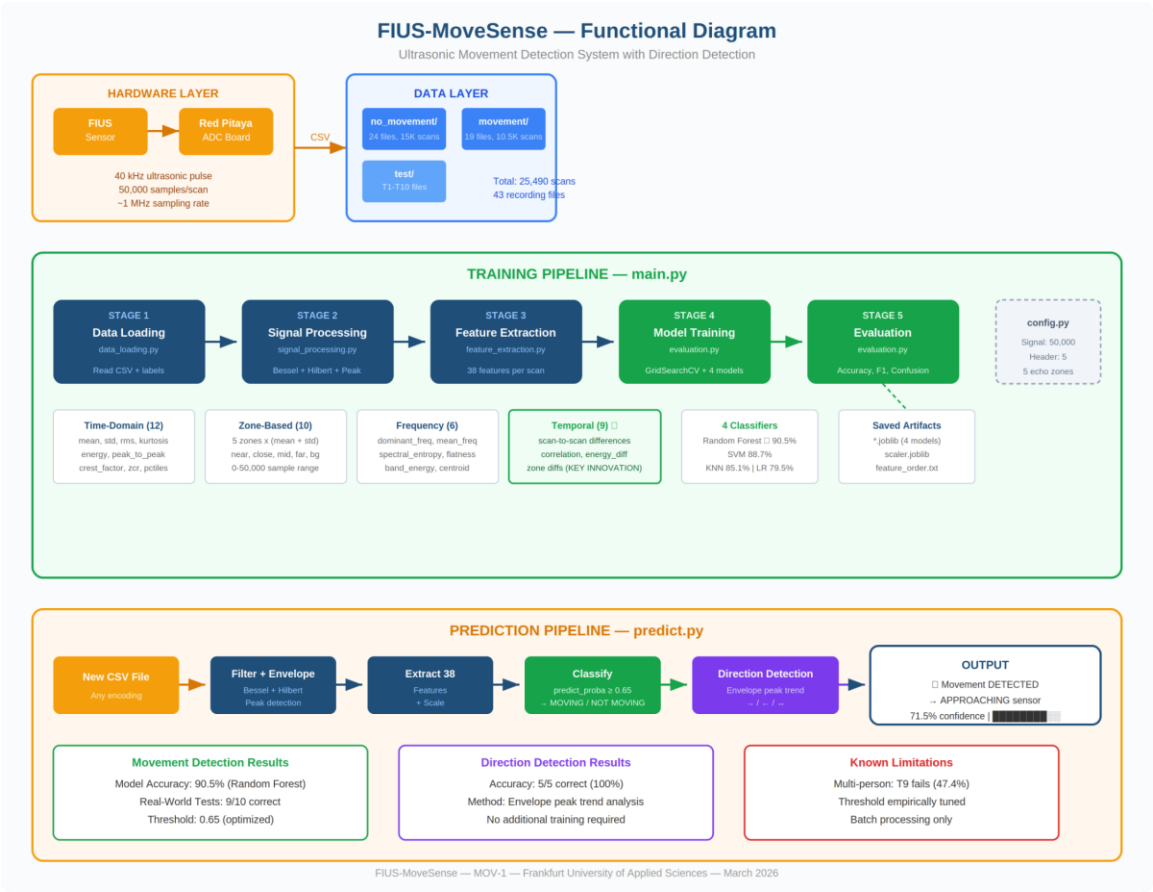
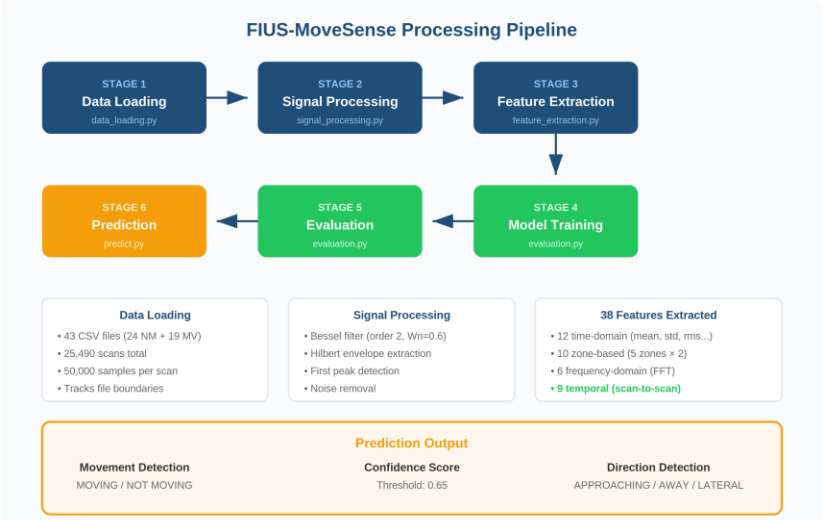


Figure 1. Functional architecture of the FIUS-MoveSense movement detection system.

Processing Pipeline



Model Performance

Four supervised machine learning models were evaluated: Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression. Among these models, Random Forest achieved the best performance with an accuracy of approximately 90.5%, precision of 85.6%, recall of 92.6%, and an F1-score of about 0.89. Experimental evaluation using ten real-world test scenarios demonstrated that the system successfully detected movement in nine out of ten cases.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	90.53%	85.56%	92.63%	88.96%
SVM	88.68%	83.50%	90.34%	86.79%
KNN	85.09%	80.02%	85.01%	82.44%
Logistic Regression	79.52%	73.09%	79.54%	76.18%

Key Achievements

- Development of a complete ultrasonic sensing and machine learning pipeline.
- Successful integration of FIUS ultrasonic sensor with the Red Pitaya platform.
- Extraction of 38 meaningful signal features for movement classification.
- Achieved approximately 90% detection accuracy using the Random Forest classifier.
- Implementation of movement direction detection using echo peak shift analysis.

Limitations

Despite promising results, several limitations remain. The dataset was collected under controlled experimental conditions and may not capture all real-world environments. Multi-person movement scenarios remain difficult to detect accurately. Additionally, the current system processes data in batch mode rather than real-time streaming.

Future Work

Future improvements may include collecting a larger and more diverse dataset, implementing real-time streaming analysis, improving multi-person detection capability, and integrating distance estimation techniques using echo peak positions. These enhancements would increase the robustness and practical applicability of the system.

Final Conclusion

In conclusion, the FIUS-MoveSense system demonstrates that ultrasonic sensing combined with machine learning can effectively detect movement in indoor environments. The results confirm that meaningful signal features extracted from ultrasonic echoes allow reliable classification of motion patterns. With an accuracy of around 90% and successful detection in most real-world tests, the proposed approach shows strong potential for applications in intelligent sensing, automation systems, and smart environments.